

Course: B.Tech.

Subject: Mathematics-IV

Max. Marks: 20

Semester: IV

Branch: EE/MME/ME

Subject Code: MTH 241

Max Time: 90 Minutes

Note: Attempt all the questions. Usual notations are used.

1	A company makes three products X, Y, and Z out of three raw materials A, B, and C. The number of units of raw materials required to produce one unit of product is as given in the following table: <table> <tr> <th></th><th>X</th><th>Y</th><th>Z</th></tr> <tr> <td>A</td><td>1</td><td>2</td><td>1</td></tr> <tr> <td>B</td><td>2</td><td>1</td><td>4</td></tr> <tr> <td>C</td><td>2</td><td>5</td><td>1</td></tr> </table> The unit profit contribution of the products X, Y, and Z are Rs. 40, 25, and 50, respectively. The number of raw materials available are 36, 60, and 45 respectively. - By formulating the linear programming problem (LPP), find the optimal product mix that will maximize the total profit. - Find the dual of the above LPP and apply the principal of duality to find its solution from the simplex table of the above primal problem.		X	Y	Z	A	1	2	1	B	2	1	4	C	2	5	1	5																																	
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2	Solve the following LPP by dual simplex method to find the optimal solution: $\text{Min } Z = 2x_1 + 3x_2 + x_3$ Subject to the constraints: $2x_1 + x_2 + 2x_3 \geq 3$; $3x_1 + 2x_2 + x_3 \geq 4$, $x_1, x_2, x_3 \geq 0$	5																																																	
3	A company needs 30, 25, 15, and 10 million litres of oil at their four sites D_1, D_2, D_3, and D_4. The oil can be transferred from three plants P_1, P_2, and P_3 where 40, 30 and 20 million litres of oil available, respectively. Costs of transporting one million litres of oil from plants to the four sites are given in the following table: <table> <tr> <th></th><th>D_1</th><th>D_2</th><th>D_3</th><th>D_4</th></tr> <tr> <td>P_1</td><td>12</td><td>7</td><td>10</td><td>10</td></tr> <tr> <td>P_2</td><td>10</td><td>8</td><td>12</td><td>10</td></tr> <tr> <td>P_3</td><td>14</td><td>12</td><td>9</td><td>14</td></tr> </table> Solve the transportation problem for the minimum cost.		D_1	D_2	D_3	D_4	P_1	12	7	10	10	P_2	10	8	12	10	P_3	14	12	9	14	5																													
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4	Six jobs, J_1, J_2, J_3, J_4, J_5, and J_6 need to be assigned to six available machines, M_1, M_2, M_3, M_4, M_5, and M_6. However, machine M_5 is incompatible for job J_4, and machine M_5 cannot process job J_6. The cost associated with each job for the respective machines is provided in the following table. Determine the optimal assignment of jobs to machines while considering these constraints. Additionally, analyze whether the optimal assignment is unique. If multiple optimal solutions exist, identify an alternate assignment. <table> <tr> <th></th><th>M_1</th><th>M_2</th><th>M_3</th><th>M_4</th><th>M_5</th><th>M_6</th></tr> <tr> <td>J_1</td><td>20</td><td>23</td><td>18</td><td>10</td><td>16</td><td>20</td></tr> <tr> <td>J_2</td><td>50</td><td>20</td><td>17</td><td>16</td><td>15</td><td>11</td></tr> <tr> <td>J_3</td><td>60</td><td>30</td><td>40</td><td>55</td><td>8</td><td>7</td></tr> <tr> <td>J_4</td><td>6</td><td>7</td><td>10</td><td>20</td><td>—</td><td>9</td></tr> <tr> <td>J_5</td><td>18</td><td>19</td><td>28</td><td>17</td><td>60</td><td>—</td></tr> <tr> <td>J_6</td><td>9</td><td>10</td><td>20</td><td>30</td><td>40</td><td>55</td></tr> </table>		M_1	M_2	M_3	M_4	M_5	M_6	J_1	20	23	18	10	16	20	J_2	50	20	17	16	15	11	J_3	60	30	40	55	8	7	J_4	6	7	10	20	—	9	J_5	18	19	28	17	60	—	J_6	9	10	20	30	40	55	5
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